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ABSTRACT

Twenty-nine samples collected from an 80-m section of the Trenton Limestone give almost vertical directions of natural remanent magnetization (NRM), but these become nearly horizontally directed after magnetic cleaning ($D = 173°$, $I = 18°$, $\alpha_{95} = 5°$). The result is confirmed by thermal demagnetization experiments, and the consequent low latitude suggested for the time of deposition is supported by the paleoclimatic evidence. Samples collected from a penecontemporaneous slump bed demonstrate that both the NRM and cleaned directions are postdepositional. Magnetic-anisotropy measurements show good agreement between the mean principal axis of maximum susceptibility and the cleaned remanence direction, suggesting that the alignment of the long axes of magnetic grains was controlled by the ancient magnetic field direction. This can only have arisen through a postdepositional precompaction in situ rotation of the magnetic grains, a process originally suggested by Irving and Major (1964) from laboratory deposition experiments. This study represents the first positive demonstration that such a process occurs naturally.

The Ordovician paleomagnetic pole position for North America deduced from these rocks (lat 36° S., long 66° W.) is in complete conflict with the predictions of the remagnetization hypothesis, which requires near-polar conditions at that time. The result, however, is in excellent agreement with that obtained from igneous rocks of similar age from the Front Range of Colorado and shows that the Ordovician pole position lies close to the late Paleozoic position. A consequence of this conclusion is the existence of an Ordovician proto-Atlantic Ocean, as has been suggested independently from geological evidence.

HISTORY OF THE REMAGNETIZATION HYPOTHESIS

Controversy has existed for a number of years over the interpretation of the paleomagnetism of lower Paleozoic rocks of North America and Europe. This arose from studies of the paleomagnetism of the Midland Valley lavas of Scotland (Stubbs, 1958; Creer and Embleton, 1967; McMurry, 1970) and from thermal demagnetization experiments on the Lower Old Red Sandstone of the Anglo-Welsh cuvette (Chamalaun and Creer, 1964). Results from these studies gave consistent but quite different pole positions from the previously accepted Devonian pole position for Europe (Irving, 1964). Arguing that the Midland Valley lavas were the only igneous rocks studied in either the Russian platform or western Europe and that the extensive studies of red beds in the USSR did not include thermal demagnetization experiments, Creer (1964, 1967, 1968a) accepted the pole from these British rocks as being the record of the true Lower Devonian pole position for Europe. All other Devonian pole positions from Europe were regarded as being records of late Paleozoic (Permian-Carboniferous) remagnetization in the rocks studied. Careful thermal demagnetization studies of Silurian-Devonian red beds from Norway (Storetvedt and Gjellestad, 1966; Storetvedt and others, 1968; Lie and others, 1969) have failed to detect the pole position determined from rocks of the British Isles. Storetvedt (1968) has, therefore, argued that it is the latter that are anomalous rather than the former.

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When the Paleozoic pole positions for Europe and North America are plotted on the traditional reconstructions of Pangaea made by Wegener (1929) and du Toit (1937), Creer (1964, 1967, 1968a) noted that the Devonian poles for the British isles coincided very well with certain lower Paleozoic poles from North America but not with others. The former included some old NRM measurements by Graham (1954, 1956) on the Trenton Limestone (Ordovician) and Onondaga Limestone (Lower Devonian) while the latter were largely results from red beds including especially the Bloomsburg Formation (Silurian) upon which extensive thermal demagnetization experiments had been carried out (Irving and Opdyke, 1965; Roy and others, 1967). Creer (1967, 1968a), therefore, proposed that those red beds, like their European counterparts, had also been extensively remagnetized in the late Paleozoic.

In order to explain why lower Paleozoic rocks from parts of Laurasia were remagnetized and rocks of the same age from Gondwanaland were not, Creer (1968b) pointed to the different latitude belts occupied by each of these supercontinents in the late Paleozoic. Laurasia occupied low latitudes while Gondwanaland occupied high latitudes, and it was supposed that many of the lower Paleozoic rock formations used for paleomagnetic purposes had subsequently occupied low latitudes in the late Paleozoic, which led to the formation of secondary iron oxides. Lateritization, one of the relevant chemical reactions, is confined to the tropical zone at the present time, so Creer (1968b) argued that the broad zone of low latitudes across Laurasia in the Permian-Carboniferous produced an environment favorable to the formation of red beds on a wide scale, never since repeated. The hypothesis put forward by Creer (1968b) has been generally referred to as the *remagnetization hypothesis.*

As a test for the remagnetization hypothesis, Creer (1968b) predicted that lower Paleozoic rocks from regions which then occupied high paleolatitudes would not have been remagnetized. He then noted that lower Paleozoic rocks from Siberia gave poles that agreed with those from the British Isles when plotted on a reconstruction of Pangaea. A consequence of this agreement was that Siberia and the Russian platform were a single unit during the early Paleozoic. Analysis of the geology of the Uralides in terms of plate tectonics suggests the Urals represent a collision zone between the formerly separated Russian and Siberian platforms (Hamilton, 1970). This view is perfectly consistent with the straightforward conclusion that the differences in pole positions from lower Paleozoic rocks on either side of the Urals are real and not due to the remagnetization hypothesis. This test is, therefore, inconclusive.

Later, Creer (1969) interpreted results from the "intermediate" directions of magnetization found by Roy and others (1967) from the Bloomsburg Formation as the first evidence from North America relevant to the remagnetization hypothesis. Creer claimed that the thermally cleaned normal and reversed directions, regarded by Roy and others as the primary Silurian directions, are records of late Paleozoic remagnetization, while a third steeply magnetized group of "intermediate" directions represents the true Silurian field direction. Roy and others (1967) regarded these directions as arising from the incomplete cleaning of certain samples even at temperatures close to the Curie point of hematite. Again, this argument is inconclusive and the evidence circumstantial. Subsequently, Creer (1970) persisted with the view that the only poles representative of the lower Paleozoic of North America are those determined from NRM directions in the Trenton and Onondaga Limestones and from the "intermediate" Bloomsburg directions. These give a mean pole at 16° N., 76° W. compatible with a common polar wander curve for all continents and the existence of Pangaea. McElhinny and Briden (1971), however, pointed out the "magnetic age" assigned to these poles (Silurian-Devonian) is incompatible with paleoclimatic evidence from southern Africa. The Ordovician-Silurian tillite of the Cape System would be placed within 30° of the equator on Creer's model but very close to the south pole on a model that rejects the remagnetization hypothesis.

The conclusive test of the remagnetization hypothesis in North America, therefore, lies in a thorough paleomagnetic investigation of either the Trenton or Onondaga Limestones. Neither of these are red beds and their magnetization is supposedly unaffected by the remagnetization hypothesis. In this respect, it is worth noting that a recent study of Cambrian-Ordovician igneous rocks from the Front Range of Colorado (Larson and Mutschler, 1971) gave an apparently anomalous pole close to upper Paleozoic poles from North America, even after all standard demagnetiza-

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tion procedures suggested that the cleaned remanence was primary.

In order to test the hypothesis directly it was decided to resample the section of the Trentonian Series that Graham (1954) had originally studied.

Thirty hand samples were collected from the Denley Limestone and lowermost Steuben Limestone of the Trentonian Series of Ordovician age exposed at Trenton Falls section, New York (Kay, 1968). An attempt was made to sample at intervals throughout the section with preference being given in the sampling to the more sparsely fossiliferous beds; however, highly fossiliferous sediments were included in the suite of samples. A majority of the samples (20) were taken from the Poland-Russia members of the Denley Limestone while nine were obtained from the Rust member of the Denley Limestone and the lowermost Steuben Limestone, three of these being sampled from a highly contorted slump structure at the top of the section in the spillway of the High Dam. The total section sampled was about 70 m (230 ft) which is the approximate thickness sampled by Graham.

The Denley Formation at Trenton Falls is composed of sparsely to highly fossiliferous, variably shaly calcarenite and calcisiltite with intercalated metabentonites. The beds are essentially flat lying with no evidence of later deformation or deep burial.

The section sampled above is the same as that sampled by Graham in 1950 and reported on in 1954 and 1956 (Graham, 1954, 1956). Graham's collection consisted of 35 samples spread throughout 70 m (230 ft) of section at Trenton Falls and 10 samples from the slumped beds at the top of the section. The NRM directions of magnetization are well grouped and steeply inclined toward the south. Graham did not publish the original NRM parameters. Cox and Doell (1960) in their review determined the declination and inclination of each point from the figure originally published by Graham. From this they calculated a mean declination of 179° with inclination of +82°, $\alpha_{95} = 5°$. Graham demonstrated that there is no difference between the NRM directions of the slumped beds and the undistorted beds beneath, showing the NRM directions observed must have been acquired after the slumping occurred.

PALEOMAGNETIC MEASUREMENTS

The samples were cored and sliced in the laboratory into 2.5 cm × 2.5 cm cylinders, between two and four specimens being prepared from each sample depending on the availability of material. One sample was unsuitable for coring and hence only 29 samples were used. All paleomagnetic measurements were made on a Digico computerized 7 Hz fluxgate spinner magnetometer, in which a Fourier analysis is made of the output waveform to determine the "in-phase" and "quadrature" components perpendicular to the spin axis for each of six spin positions (Molyneux, 1972). NRM intensities ranged from 1 to 9 × 10^{-6} emu cm^{-3} so that 28 spins were always analyzed in each position. At this level it was found relatively easy to measure specimens whose intensities ranged down to 1 or 2 × 10^{-7} emu cm^{-3}.

NRM directions of 29 samples (97 specimens) were well grouped about a nearly vertical direction (Table 1; Fig. 1), very close to that obtained by Graham (1954) from a similar suite of samples. Five pilot specimens were partially demagnetized in alternating magnetic fields in increasing steps of 50 oersteds peak value (Fig. 2a), using an apparatus similar

TABLE 1. PROGRESSIVE ALTERNATING FIELD DEMAGNETIZATION OF SAMPLES OF TRENTON LIMESTONE

Treatment	Samples (specs)	D	I	R	k	α_{95}
NRM	29(97)	153.5	88.1	28.635	76.7	3.0
100 oersteds	29(29)	176.1	54.4	27.386	17.3	6.6
200 oersteds	29(29)	172.4	31.9	25.745	8.6	9.6
Cleaned result (200-300 oe)	29(91)	173.1	18.5	27.943	26.4	5.2

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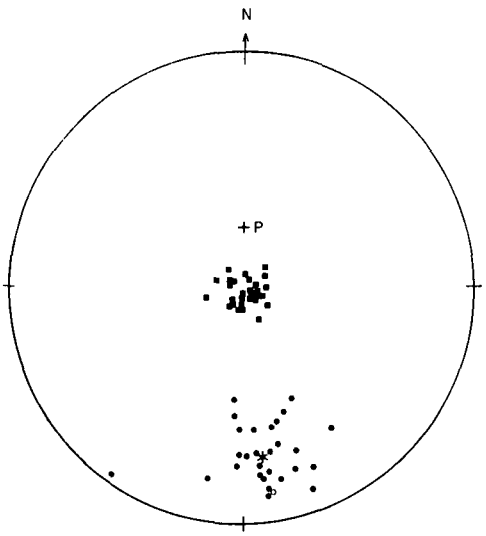


Figure 1. Directions of magnetization for all specimens before (filled squares) and after (filled circles) a.f. demagnetization in peak alternating fields of 200 oersteds. The present dipole field is indicated by +P. The mean direction of magnetization after a.f. demagnetization is shown by the star.

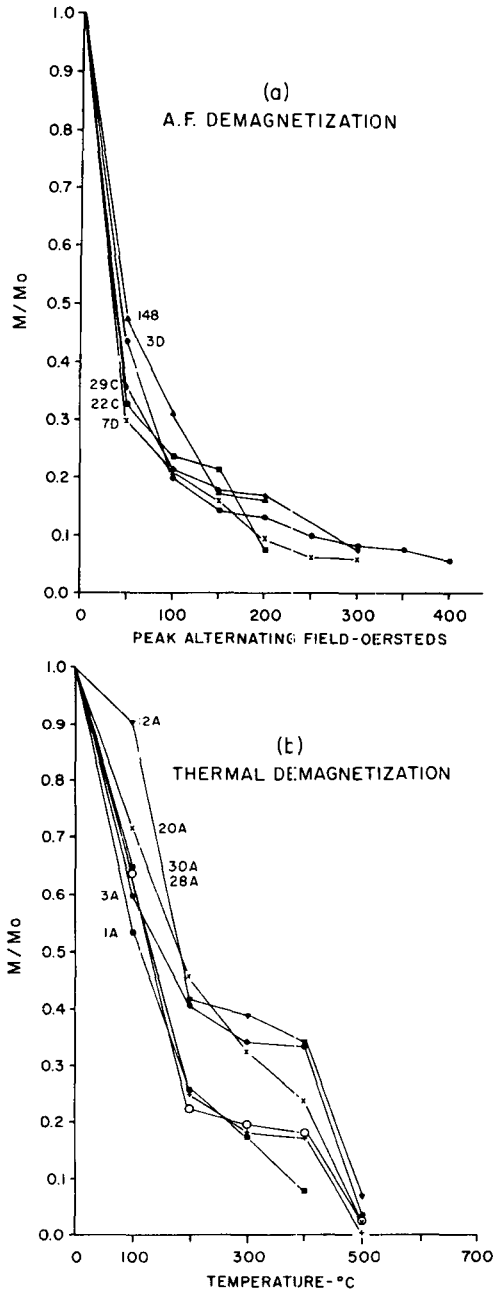


Figure 2. Alternating field (a) and thermal demagnetization curves (b).

to that described by McElhinny (1966) but with the field being raised and lowered with a G. E. Inductrol. In a peak field of 50 oersteds, the intensities of all specimens had dropped to less than half their NRM value and by 200 oersteds less than 20 percent remained. Directions changed progressively to become southerly and nearly horizontal. A large component of magnetization directed along the present field direction was being progressively removed, requiring in the region of 200 to 300 oersteds for magnetic cleaning to be completed. The specimens, however, were becoming weakly magnetized (intensities ranged from 2 to 9 $\times$ 10^{-7} emu cm^{-3}), and it was decided to partially demagnetize one specimen from each sample in peak fields of 100, 200, and 300 oersteds. From these measurements it was determined that the optimum cleaning field varied between 200 and 300 oersteds. Progressive demagnetization is shown in Figure 3 for all 29 samples, the final point representing the cleaned direction using 91 specimens. Directions of the cleaned samples are illustrated in Figure 1 and the progressive demagnetization characteristics are listed in Table 1.

Six specimens were saved for an investigation of the thermal demagnetization characteristics. The specimens were heated and cooled in a field of less than 10 gammas. Temperature calibration was a little uncertain, but certainly accurate to better than 20°C. The progressive thermal demagnetization characteristics are shown in Table 2. The thermal demagnetization curves (Fig. 2b) suggest the same as the

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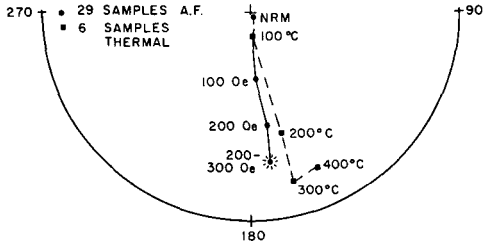


Figure 3. Progressive change in mean direction of magnetization with increasing alternating fields in 29 samples given by the filled circles, compared with a similar change in mean direction with increasing temperatures in 6 samples shown by the filled squares.

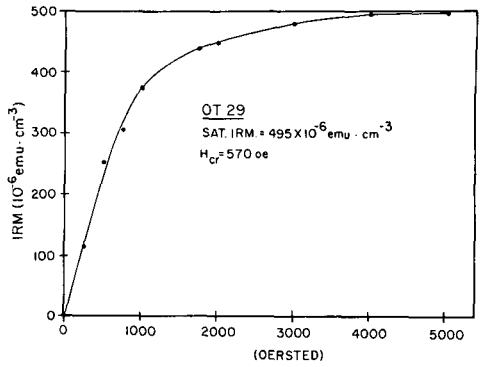


Figure 4. Saturation remanent magnetization of sample OT29.

a.f. curves (Fig. 2a), that a large viscous component is removed up to 200°C, while the directions are progressively southerly with shallow inclinations (Fig. 3). The thermal demagnetization curves flatten out between 200°C and 400°C, the best grouping being at 300°C and giving a mean direction close to that found through magnetic cleaning (Fig. 3). At 500°C, the intensities of all specimens drop to the lower limit of the magnetometer; the sharp drop from 400°C suggests that this is close to the Curie temperature of the magnetic mineral. This mineral thus appears to be a titanomagnetite, consistent with the observation that these limestones contain volcanic ash. Reflection microscopy was attempted but magnetic particles were too finely divided to identify. The curves of isothermal remanent magnetization (IRM) (Fig. 4) show a rapid increase up to 2 kOe with saturation occurring around 4 kOe. These characteristics exclude hematite and are consistent with the main magnetic mineral being magnetite or a titanomagnetite.

ORIGIN OF THE MAGNETIZATION

Both the NRM and cleaned directions of magnetization shown in Table 1 refer to the present-day horizontal. If corrections are made to those samples collected from the slump beds so as to refer the magnetizations to the bedding plane, then for both the NRM *and* cleaned directions the slump magnetizations become more scattered, diverging from the main group by up to 45°. Graham (1954) also noted this point for the NRM directions, concluding that the NRM was postdepositional. It has been shown (Rees, 1965) that, in favorable circumstances, the measurement of anisotropy of magnetic susceptibility may be used to estimate the preferred grain orientation of sediments. Grains of magnetite or titanomagnetite are most easily magnetized along their longer dimensions and so any rock containing a small proportion of such grains has directional differences in susceptibility that define a

TABLE 2. PROGRESSIVE THERMAL DEMAGNETIZATION OF SIX SPECIMENS OF TRENTON LIMESTONE

Temperature	D	I	R	k	α_{95}
NRM	148.2	86.4	5.956	113.1	6.3
100°C	176.9	76.8	5.766	21.3	14.8
200°C	166.5	28.0	5.614	12.9	19.3
300°C	165.9	9.7	5.952	104.7	6.5
400°C	157.5	12.2	5.808	26.0	13.3

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magnetic fabric, reflecting preferred orientation of their long and short axes. The anisotropy of magnetic susceptibility was, therefore, determined for one specimen in each sample using a PAR-SM1 spinner magnetometer. The specimen is spun in a steady field of 4.86 oersteds at 52.5 Hz, one-half the frequency used for remanence measurements. This produces a time-varying signal frequency of twice the spinning frequency and of magnitude proportional to the applied field and to the anisotropy in the plane at right angles to the spinning axis. The method is discussed by Noltimier (1967) but in this case a six-spin technique and computational procedure developed by W. Lowrie (personal commun.) were used. Signals were near the lower limit of measurement on the magnetometer.

The method of measurement and computation of the anisotropy of susceptibility estimates a susceptibility in the form of a second-rank symmetric tensor K (Granar, 1958), which relates the magnetization J_i induced in a specimen by a field H:

$$J_i = K_{ij} H_j .$$

The tensor describes a representative orientation for all the magnetic grains in a sediment and gives some indication of the perfection of alignment. The susceptibility can be completely specified by six quantities, three relating to the magnitudes of its principal axes K_{max}, K_{int}, and K_{min} and three relating to their directions which are orthogonal. In sediments the most important directional property is the principal axis of maximum susceptibility, which gives the estimated direction of preferred long-axis orientation for each specimen. The directions of the principal axes of maximum and minimum susceptibility determined for one specimen from each of the 29 samples are plotted in Figure 5. The average directions of these principal axes (maximum A, minimum C) are calculated in Table 3, and are shown in Figure 5 together with a circle of confidence determined on the assumption of a Fisher (1953) distribution. The total anisotropy (A_n) for each specimen is given by

$$A_n = \frac{(K_{max} - K_{min})}{K_{int}}$$

following Rees (1966) and varies from 0.8 to 4.6 percent with an average value of 2.2 percent and median value of 1.8 percent. We note, however, that the directions of the principal axes of maximum susceptibility differ

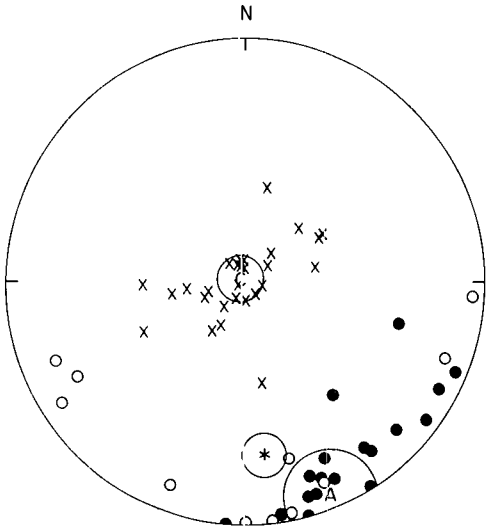


Figure 5. Axes of maximum (circles) and minimum (crosses) susceptibility. The mean direction of maximum susceptibility is indicated by A and of minimum susceptibility by C. The mean direction of magnetization is given by the star for purposes of comparison. The circles are the α_{95}'s.

from those determined by Graham (1966). We can find no explanation for this except to say that quite different techniques of measurement were used in each case.

King and Rees (1966) have shown that for magnetic particles being deposited in still water the gravitational couple tending to control the long-axis orientation will, in general, outweigh the orienting effects of the magnetic field. Since elongated particles will tend to lie in the bedding plane, this means that, for particles deposited in still water onto a horizontal bed, the principal axis of maximum susceptibility will tend to be randomly directed in the bedding plane as has been demonstrated experimentally by Rees (1966). Preferred orientation of the principal axis of maximum susceptibility will, however, occur if the bed is sloping (in the direction of dip) as shown by Rees (1966) or if water currents are present (in the direction of flow) as shown by Rees (1961) and Hamilton (1963). Although the principal axes of maximum susceptibility in specimens of Trenton Limestone lie in the horizontal plane and have a preferred southerly orientation, this axis is not related to the bedding plane in the slump beds. The slumping appears to have had no effect on the preferred

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TABLE 3. AVERAGE DIRECTIONS OF PRINCIPAL AXES OF MAXIMUM AND MINIMUM SUSCEPTIBILITY FOR THE TRENTON LIMESTONE

Axis	N	D	I	R	k	α_{95}
Maximum (A)	29	158	3	23.31	4.9	13.4
Maximum (C)	29	310	88	26.36	10.5	8.6

orientation of the maximum-susceptibility axis. From this observation it must be concluded that the preferred orientation of the long axes of the magnetic grains took place *after* slumping and is not related to conditions (such as water currents) prevailing at the time of deposition. In Figure 5 the close correspondence between the average principal axes of maximum susceptibility and the direction of the cleaned remanence suggests that orientation of the long axes was controlled by the ancient magnetic field. This correspondence can only arise for multidomain grains, because the moments of single-domain grains cannot be increased along their long axes with a 4.86 oersted field (the moments can, of course, be increased perpendicular to the long axes). If the magnetic mineral is a titanomagnetite, then the alternating field demagnetization characteristics (poor stability beyond 300 or 400 oersteds) support the view that the grain sizes lie mainly in the multidomain region (Evans and McElhinny, 1969). The experiments of Løvlie and others (1972) on deep-sea sediments suggest that a range of grain sizes will be present including both single and multidomain grains.

In order to test for the presence of multidomain grains we have applied the test suggested by Lowrie and Fuller (1971). The test uses the fact that the stability against alternating field demagnetization of thermal remanent magnetization (TRM) carried by multidomain magnetite increases with increasing strength of the inducing field, a trend opposite to that of single domain particles. Because heating specimens will undoubtedly cause chemical alteration, we have compared the a.f. demagnetization characteristics of the anhysteretic remanent magnetization (ARM) acquired in a steady field of one oersted with those of the saturation IRM as shown in Figure 6. The saturation IRM is more resistant to a.f. demagnetization than the low-field ARM as would be expected if multidomain grains predominate. While it is not yet known the extent to which a low-field ARM gives the same a.f. demagnetization characteristics as a low-field TRM, the close correspondence between the a.f. curves of NRM and ARM of rocks dredged from the Mid-Atlantic Ridge strongly suggests this to be the case (Park and Irving, 1970). The results of the test provide some support for the interpretation of the anisotropy measurements.

If the orientation of the long axes of the magnetic grains was controlled by the ancient magnetic field, this cannot have occurred through tectonic deformation at a later date, since the rocks are undisturbed and, in any case, deformation of the magnetic fabric is often associated with other signs of deformation (Graham, 1966). Therefore, it must be concluded that orientation of the magnetic grains took place penecontemporaneously with deposition, but after slumping and before

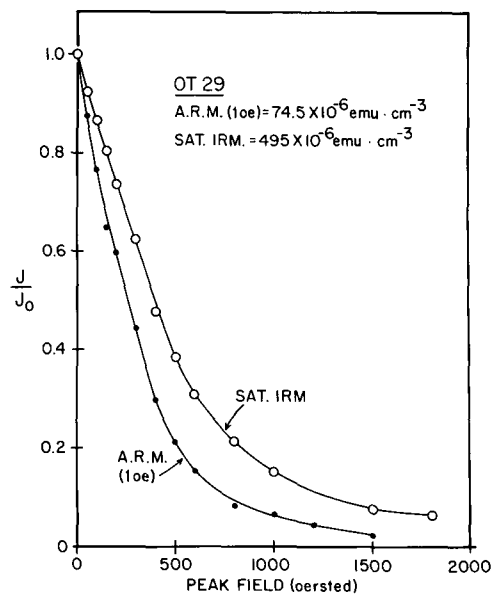


Figure 6. Alternating field demagnetization of saturation IRM and ARM of specimen OT29.

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lithification and compaction. This corresponds precisely with the postdepositional detrital remanent magnetization (DRM) observed by Irving and Major (1964) in their laboratory depositional experiments. Failure to observe any "inclination error," as had been established by King (1955) from laboratory experiments, led Irving and Major (1964) to interpret their observations as indicating the possibility of postdepositional rotation of magnetic grains. It appears that the present investigation is the first reported field demonstration of the postdepositional DRM process proposed by Irving and Major (1964). The failure to observe systematic inclination error in deep-sea sediments (Opdyke and Henry, 1969) and the work of Løvlie and others (1972) on their magnetic properties suggest that this process also operates in the case of these sediments.

LOWER PALEOZOIC POLES FOR NORTH AMERICA

Table 4 and Figure 7 summarize all the most reliable lower Palaeozoic poles that have been determined for North America using cleaning techniques. A pole calculated from the NRM measurements in the present study lies at 40° N., 73° W., quite close to that determined originally from Graham's (1954) data by Cox and Doell (1960). The pole derived from the cleaned results agrees well with that determined by Larson and Mutschler (1971) from Cambrian-Ordovician igneous rocks in the Front Range of Colorado. They differ widely from some results on Ordovician sedimentary rocks from eastern Newfoundland determined by Deutsch and Rao (1970), but this is not surprising in view of the complex tectonic history of the island (Bird and Dewey, 1970). Consequently, all other results from Newfoundland are not included in Table 4. Mean poles for the upper Paleozoic are also given in Table 4 and plotted in Figure 7. The Trenton pole and that for the Cambrian-Ordovician igneous rocks lie close to the upper Paleozoic poles as would be expected on the remagnetization hypothesis. We note, however, that not only has the magnetization of the Trenton Limestone been demonstrated to date from Ordovician times, but also neither limestones nor igneous rocks should be affected by the remagnetization hypothesis, because neither are red beds. There are now three independent determinations of lower Paleozoic pole positions that are in close agreement, and they involve three types of remanence: DRM in the Trenton Limestone (this paper); TRM in the Front Range intrusives (Larson and

TABLE 4. PALEOZOIC POLES FOR NORTH AMERICA (AS PLOTTED IN FIGURE 7)

Rock Unit	Age	Pole Position	α_{95}(dp,dm)	Reference†
Lamotte Fm. (Group IIa)	upper Cambrian	38° N., 12° E.	(4,8)	Al-Khafaji and Vincenz (1971)
Lamotte Fm. (Group IIb)	upper Cambrian	12° N., 25° W.	(7,11)	Al-Khafaji and Vincenz (1971)
Lamotte Fm. (Group 1)§	upper Cambrian	30° S., 7° E.	(8,9)	Al-Khafaji and Vincenz (1971)
Lamotte Fm. (Locality E)§	upper Cambrian	(32° S., 43° W.)	(10,20)	Al-Khafaji and Vincenz (1971)
Ratcliffe-Brook Fm.	lower Cambrian	10° S., 56° W.	(14,21)	Black (1964)
Wichita granites	Cambrian	2° S., 33° W.	(6,9)	Spall (1968)
Wichita Mountains§	Cambrian	(34° S., 46° W.)	(7,15)	Ku and others (1967)
Intrusives, Colorado	Cambrian-Ordovician	40° S., 69° W.	-	Larson and Mutschler (1971)
Trenton Limestone	middle Ordovician	36° S., 66° W.	(3,5)	This paper
Wabana Group	lower Ordovician*	28° S., 12° E.	(2,4)	Deutsch and Rao (1970)
Castanea Fm.	lower Silurian	21° S., 75° W.	(9,13)	Opdyke (in preparation)
Bloomsburg Fm.	upper Silurian	32° S., 78° W.	(6,10)	Roy and others (1967)
Devonian mean (4)	Devonian	29° S., 61° W.	10	8/120-1; 9/120; 10/126
Miss.-Penn mean (11)	Mississippian-Pennsylvanian	38° S., 55° W.	5	8/(95-7), 99-101, 113; 9/98 10/115, 119-20; 11/76,78
Permian mean (6)	Permian	46° S., 63° W.	7	8/88; 9/65; 10/105-6; 11/65, 67

Note: Poles in parentheses are those we consider to be late paleozoic remagnetization.
**From eastern Newfoundland*
†The numbers refer to entries in the pole Lists of McElhinny (1968a, 1968b, 1969, 1970, 1972).
§Not plotted in Figure 7, see text.

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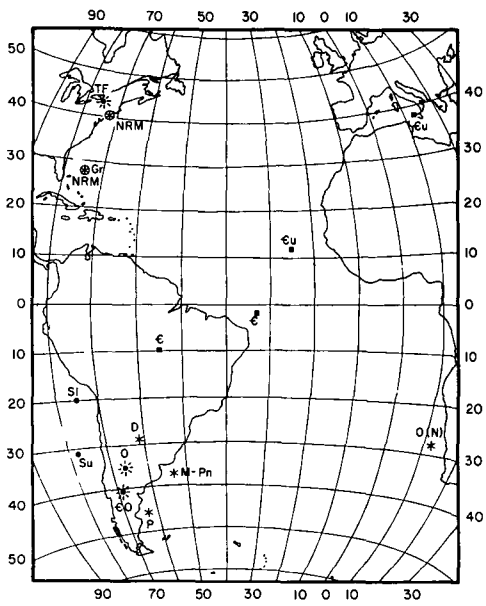
Mutschler, 1971); and chemical remanent magnetization (CRM) in the Bloomsburg red beds (Roy and others, 1967). It must be concluded that the Ordovician pole for North America lay close to its upper Paleozoic position, a conclusion that virtually discounts the basis of the remagnetization hypothesis. It now appears to be time to abandon the remagnetization hypothesis.

The interpretation of results from Cambrian rocks in North America is difficult because of the widely different poles determined by different authors (Table 4; Fig. 7). Al-Kahfaji and Vincenz (1971) found four different directions of magnetization in a study of the Lamotte Formation of Missouri. Two of these (Group IIa and IIb, Table 4) were interpreted as representing different poles during the Upper Cambrian from stratigraphic considerations; one (Group 1) was unexplained while the fourth (Locality E) was attributed to remagnetization during the Carboniferous due to the proximity of its pole to other late Paleozoic poles. A similar explanation was given by Ku and others (1967) for results from the Wichita Mountains, Oklahoma. It should be noted that neither of these results can be explained on Creer's (1968b) remagnetization hypothesis since they are either from igneous rocks or white sandstones.

Al-Khafaji and Vincenz (1971) interpreted their results together with other Cambrian results as indicating a movement of the pole in Cambrian times from a position near 40° N., 19° E. (see Fig. 7), crossing the equator at about 40° W. so as to join with the late Paleozoic poles around 40° S., 60° W. The interpretation of the Cambrian results is still speculative but our new Ordovician data together with those of Larson and Mutschler (1971) suggest that the Ordovician pole was close to its late Paleozoic position but moved to lower latitudes during the Silurian before returning again to its late Paleozoic position (Fig. 7).

ORDOVICIAN PALEOGEOGRAPHY

The paleomagnetic results presented in this paper lend strong support to the hypothesis of a proto-Atlantic Ocean (Wilson, 1966; Dewey and Bird, 1970) being in existence in Middle Ordovician time. The paleomagnetic results from the Gondwana continents would place the pole relative to these continents in northern Africa. This position for the pole is supported by the evidence of glaciation in the Ordovician of northwest Africa. If North America and northwest Africa are placed in a


90 70 50 30 10 0 10 30
50 40 30 20 10 0 10 20 30 40 50
TF
NRM
Gr
NRM
Cu
Cu
Cu

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contiguous position, which they must have occupied prior to the formation of the present Atlantic Ocean, then one would expect the Ordovician rocks of New York State to have steep inclinations. However, we have shown that North America lay in low latitudes during the Middle Ordovician. This position necessitates that a proto-Atlantic Ocean of considerable dimensions was in existence between North America and Africa and must have closed by late Paleozoic times. A position for North America in low latitudes is in agreement with available paleoclimatic evidence from North America such as the wide distribution of shallow marine carbonates and the development of evaporites in the Arctic basin (Dott and Batten, 1971). Figure 8 shows a possible reconstruction of the proto-Atlantic region for the Ordovician which satisfies the paleomagnetic data for the Ordovician.

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